Global Oil Trade

Shadow Hubs

Keerti Rawat · Muskan Khepar · Nikhil Kumar · Stiles Clements · Zan Merrill

Social Media & Network Analytics - Spring 2026

The Question

Billions of dollars in sanctioned oil still moves around the world every year. It doesn't vanish, it reroutes through intermediary countries that act as hidden middlemen in the trade network.

Use Network Analytics

1) Which countries are "**shadow hub**" (have more network centrality than their trade volume alone would predict)?

2) Which countries are the **biggest "emergent" shadow hub** over time?

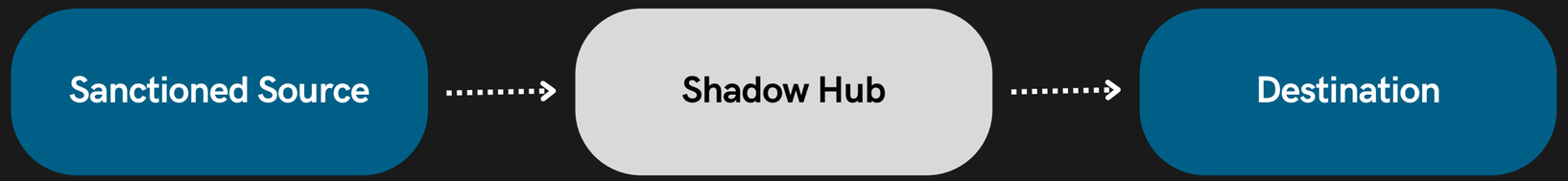
3) How does **sanctions exposure** (high OFAC footprint) relate to shadow hubs?

4) For any shadow hub, who are it's **top trade partners** in a given year?

Why It Matters

- OFAC sanctions target entities (firms, vessels, individuals), not entire countries.
- Oil trade doesn't disappear, it reroutes.
- Rerouting concentrates through a small set of intermediary countries ("shadow hubs").
- Finding these hubs gives investigators a prioritized shortlist for deeper OSINT / compliance review.

We use network structure to detect where sanctioned trade is most likely being intermediated.



Data Sources

UN Comtrade
(Country level Trade)

→ **Bilateral Trade Tables for**
- HS 2709 (Crude oil)
- HS 2710 (Non-Crude Oil)

→ **Yearly data from 2019-2024**

→ **232 Countries with 45,513 directed trade edges**

→ **Trade Values in USD**

OFAC SDN
(Sanctioned Entities)

→ **9,000+ Sanctioned Entities data**
- Individuals
- Companies
- Vessels

→ **Latest Snapshot**

→ **177 Countries signal from addresses**

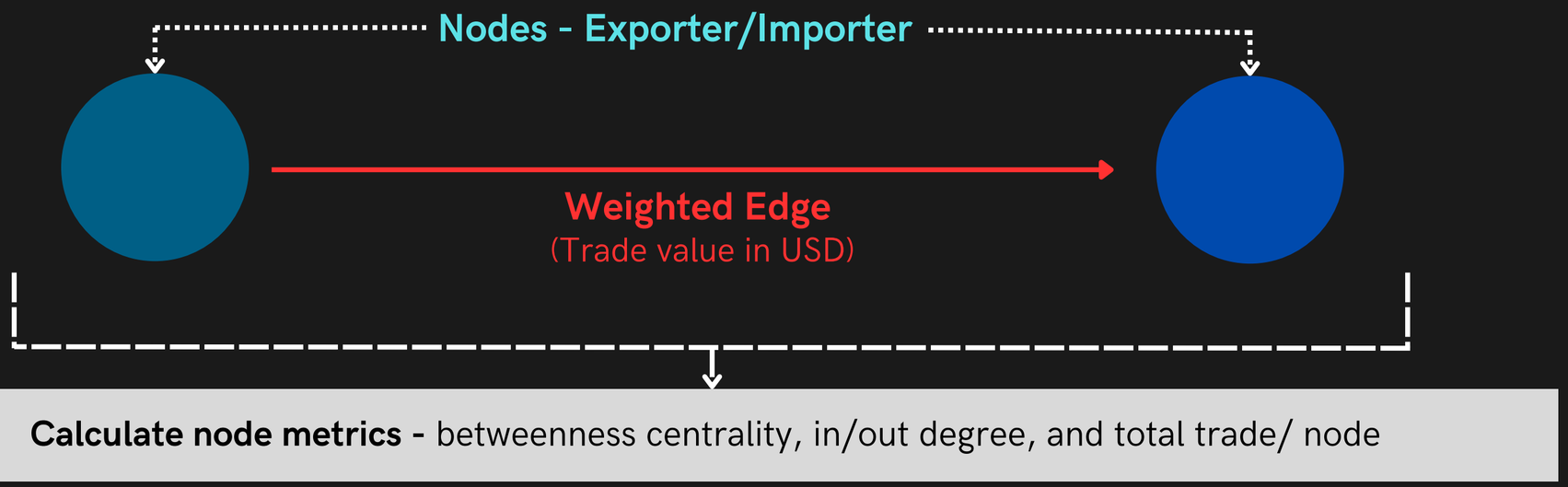
→ **Aggregated counts by ISO3**

Data Pipeline

1. **Filter and Standardize Comtrade data** by keeping only country-to-country flows with trade value and normalizing identifiers to ISO3 and cleaning unknowns
2. **Construct network tables** as an edges table (exporter_iso3 → importer_iso3, year, trade_value_usd) and a nodes table (unique country_iso3 + metadata)
3. **Parse + map OFAC SDN** by extracting each entity's primary country, normalizing country text, and converting to ISO3 (overrides + pycountry) with coverage checks
4. **Aggregate + merge OFAC** by rolling entity-level OFAC into country-level counts (ofac_entities, ofac_entity_country_links) and left-joining onto the country nodes table
5. **Compute per-year network metrics** (betweenness, degrees, total trade volume) from the directed trade graph built from edges

Building the Network

We built directed weighted graphs for each year from 2019-2024 using UN Comtrade bilateral oil flows. 232 country nodes connected by 45,513 trade edges.



Shadow Hub Detection

Methodology

Regress log(betweenness) vs log(trade volume) for per year's network

↓

Use residual as shadow hub score

↓

Residual > 0 ⇒ "more central than expected"
Residual < 0 ⇒ "less central than expected"

↓

Overlay OFAC SDN entity count (jurisdictional sanctions footprint)

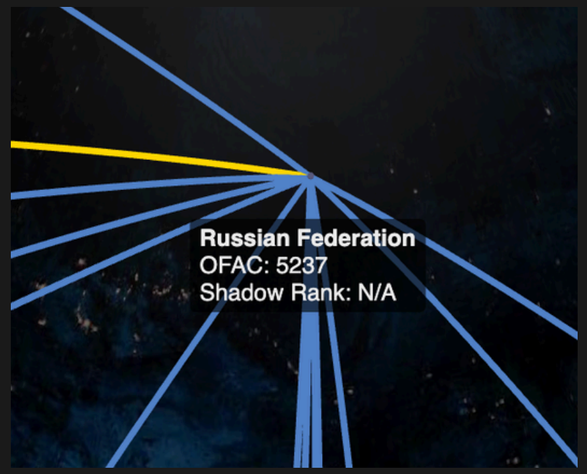
Interpretation

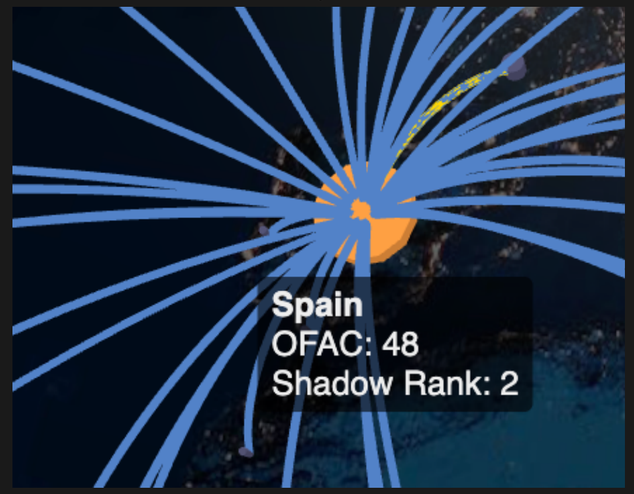
Low Residual High OFAC Sanctions-heavy, not a hub	**High Residual High OFAC** Shadow hub + sanctions footprint
Low Residual Low OFAC Baseline / typical	**High Residual Low OFAC** Unexpected intermediary

Network Analysis

1. Massive global economies break standard network anomaly models
2. 2022 Sanctions shattered direct trade, forcing complex European reroutes
3. Emerging hubs: highest change in residual shadow (From 2021 to 2024)

After 2022 sanctions

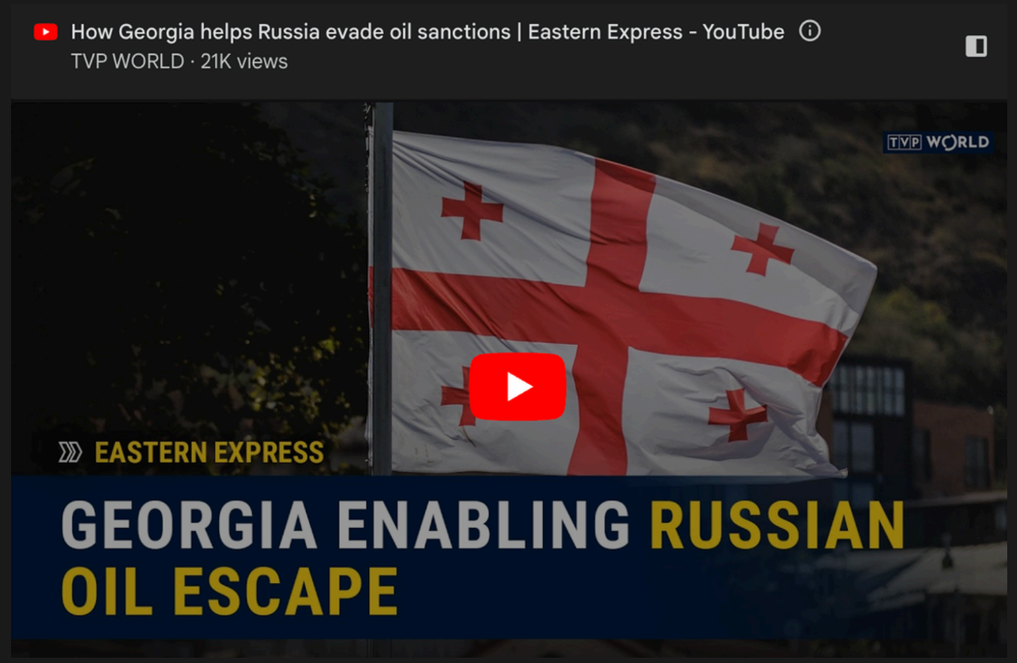




GEORGIA- the laundering syndicate

1. #1 Highest Weighted Clustering Score globally based on 2024 data
2. The Oil "Washing Machine" Effect
3. Disguise sanctioned Russian crude as a clean "Caspian blend."

Validated by a recent investigative journalism!



https://youtu.be/CwIZ8aACRSo

From Tables to a Graph DB

Neo4j AuraDB Load Pipeline — Shadow Hubs in Global Oil Trade

Step 1 — Constraints & Indexes

```cypher
CREATE CONSTRAINT country_iso3_unique IF NOT EXISTS
FOR (c:Country) REQUIRE c.iso3 IS UNIQUE;

CREATE CONSTRAINT year_unique IF NOT EXISTS
FOR (y:Year) REQUIRE y.year IS UNIQUE;
```

Step 2 — Load 232 Country Nodes via CSV (OFAC entity counts already merged)

```cypher
LOAD CSV WITH HEADERS FROM 'https://drive.google.com/...' AS row
MERGE (c:Country {iso3: row.country_iso3})
SET c.name = row.country_name,
    c.ofac_entities = toInteger(row.ofac_entities),
    c.ofac_links    = toInteger(row.ofac_entity_country_links);
```

Step 3 — Load ~45K Directed TRADE Edges via CSV

```cypher
LOAD CSV WITH HEADERS FROM 'https://drive.google.com/...' AS row
MATCH (exp:Country {iso3: row.exporter_iso3})
MATCH (imp:Country {iso3: row.importer_iso3})
CREATE (exp)-[:TRADE {
    year:            toInteger(row.year),
    trade_value_usd: toFloat(row.trade_value_usd)
}]->(imp)
```

Step 4 — Create Year Nodes + 413 SHADOW_HUB Relationships via CSV

```cypher
LOAD CSV WITH HEADERS FROM 'https://drive.google.com/...' AS row
MATCH (c:Country {iso3: row.iso3})
MATCH (y:Year {year: toInteger(row.year)})
CREATE (c)-[:SHADOW_HUB {
    betweenness, trade_total_usd, in_deg, out_deg,
    btw_expected, shadow_resid, shadow_rank
}]->(y);
```

Step 5 — Add Clustering Coefficients via batch Cypher SET The initial CSVs didn't include clustering. We computed weighted clustering coefficients (`nx.clustering(G, weight='log_weight')`) from the network analysis notebook across all 413 country-year pairs, verified against the original output, then wrote them onto the existing SHADOW_HUB relationships:

```cypher
MATCH (c:Country {iso3:'GEO'})-[s:SHADOW_HUB]->(y:Year {year:2022})
SET s.clustering_weighted = 0.46536;
-- ×413 relationships
```

Final Schema:

- **238 Nodes —** `Country` (232) + `Year` (6)
- **45,926 Relationships —** `TRADE` (~45.5K) + `SHADOW_HUB` (413)
- **Property keys on SHADOW_HUB:** betweenness, clustering_weighted, btw_expected, shadow_resid, shadow_rank, trade_total_usd, in_deg, out_deg

Neo4j Schema

Database information

Nodes (238)

* Country Year

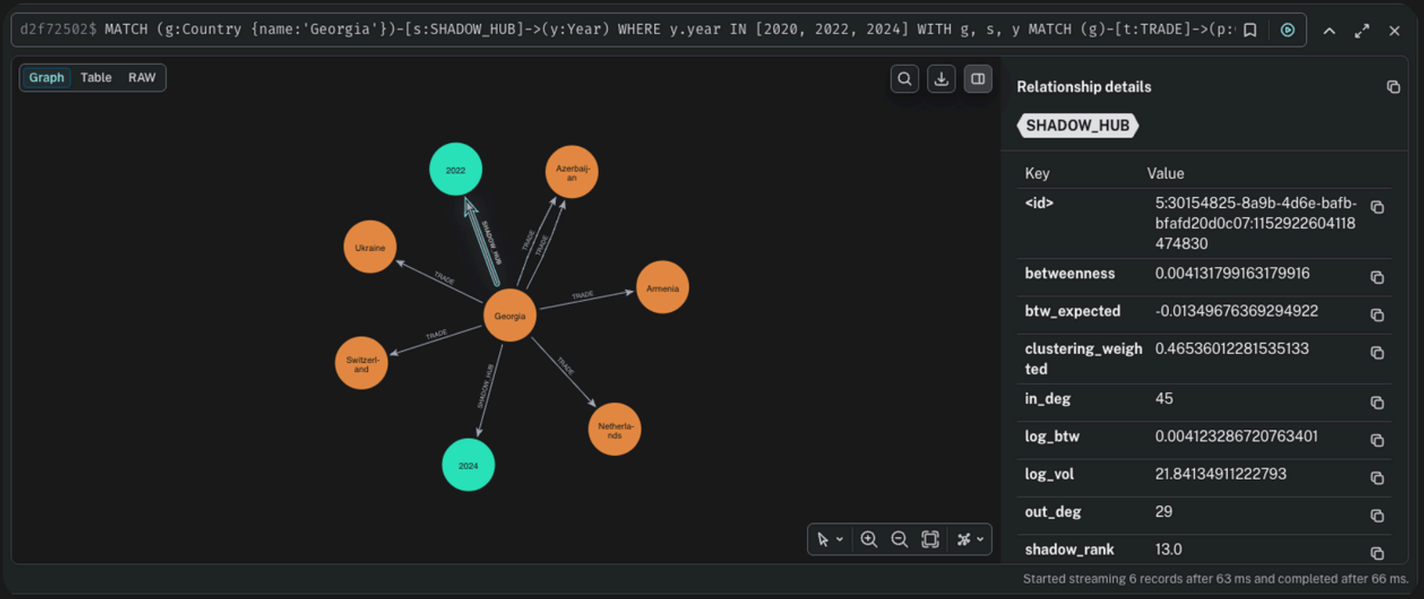
Relationships (45,926)

* SHADOW_HUB TRADE

Property keys

betweenness btw_expected
clustering_weighted comtrade_code data id
in_deg iso2 iso3 log_btw log_vol name
nodes ofac_entities ofac_links out_deg
relationships shadow_rank shadow_resid
style trade_total_usd trade_value_usd
visualisation year

Show less



GraphRAG Pipeline

Plan

- LangGraph state machine orchestrates the pipeline
- Intent Classification (7 types)
- Country / Year / top-N extraction + ambiguity resolution
- Meta & general questions routed to embedded project reference

Retrieve & Draft

- Cypher Queries + vector similarity search (hybrid retrieval)
- GPT-4o drafts cited answer with [C#] / [V#] inline citations
- General questions use embedded project reference instead of graph evidence

Evaluate & Refine

- Rule-based + LLM quality checks on draft
- Finalize / rewrite & loop (up to 2x) / ask for clarification
- Meta questions skip strict citation requirements

GraphRAG - Key Improvements

Query Understanding

- Added detection for highest betweenness queries
- Better handling of ambiguous or incomplete queries
- Added support for cross-year questions (e.g., highest residual across all years)

Answer Reliability

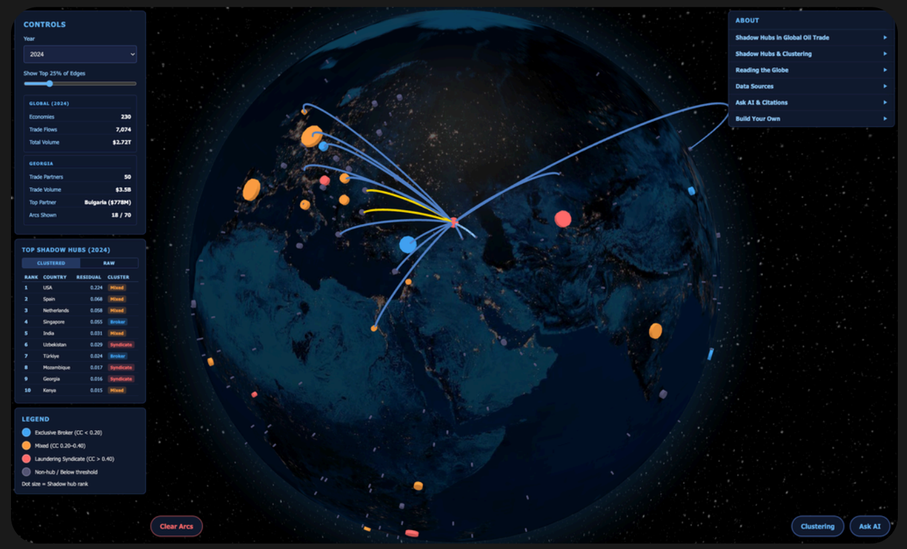
- Reduced hallucination by forcing evidence-grounded answers
- Improved evaluation stage to catch weak or unsupported answers
- Strengthened rules for interpreting shadow hub metrics

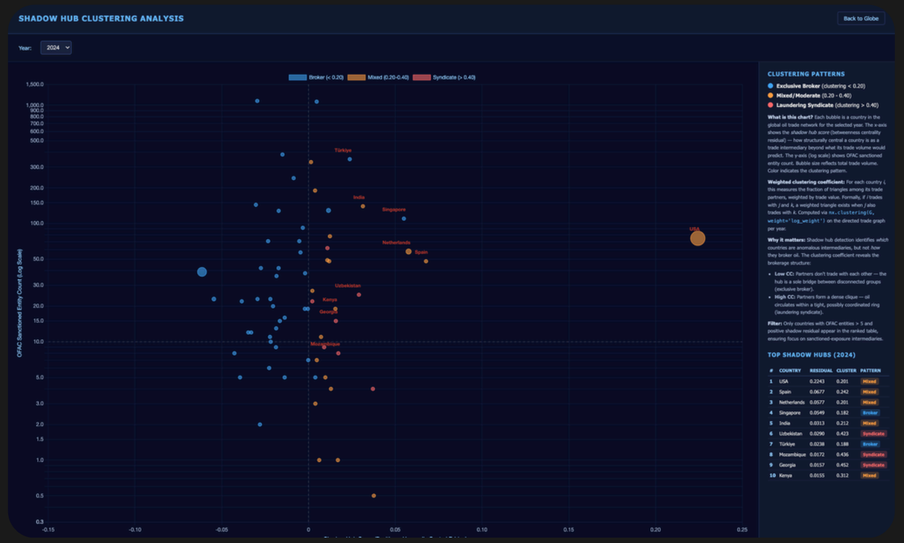
Retrieval Enhancements

- Strengthened graph-first retrieval, ensuring exact Neo4j metrics are used before vector-based evidence
- Added specialized Cypher queries for network centrality metrics and shadow hub ranking

Interactive Globe

1. Globe.gl + WebGL 3D globe where every country is a clickable point
2. Detailed annual clustering visualizations on alternate view
3. GraphRAG “Ask AI” built in: Python API backend connects the globe to the LangGraph Pipeline and Neo4j





One-Stop Insights Resource

Caveats and Next Steps

Limitations

- Shadow hubs flag structural anomalies, not definitive sanctions evasion
  - High residual = more central than trade volume predicts
- Clustering adds a second lens: "Brokers" bridge disconnected groups, "Syndicates" operate in tight trading blocs
  - Legitimate explanations exist: geography, refining capacity, trade agreements
- This is a screening tool: where to look, not what you'll find

Future Data Layers

- The current model shows where to look. Adding ground-truth data would show who is actually doing it:
  - Beneficial ownership data to trace who controls entities behind anomalous flows
  - Vessel tracking (AIS) to flag dark voyages and ship-to-ship transfers
  - Entity-level sanctions matching against SDN, EU, and UN lists
  - Financial flow data (SWIFT) to verify money flows match physical trade

Thank You

Explore the project and try it yourself at the links below

Shadow Hubs Globe
GitHub